

Program 6: Deploy an automated build pipeline using docker hub

File structure:

/Dockerfile

/package.json

Create folder >> mkdir lab6 >> cd lab6

Dockerfile

FROM alpine

Using Docker File

```
#sudo docker build -t program6 .
```

Create github personal access token

Login to github > settings > developer settings > personal access tokens > classic > give name and expiry date > check write packages and read packages

Create and copy the pat

Tag image for dockerhub

```
#docker tag program6 anantha15/program6:latest
```

Tag image for ghcr.io

```
#docker tag program6 ghcr.io/anantham016/program6:latest
```

Push to the docker hub

```
#docker login -u anantha15
```

```
#docker push anantha15/program6
```

Login to GHCR using docker

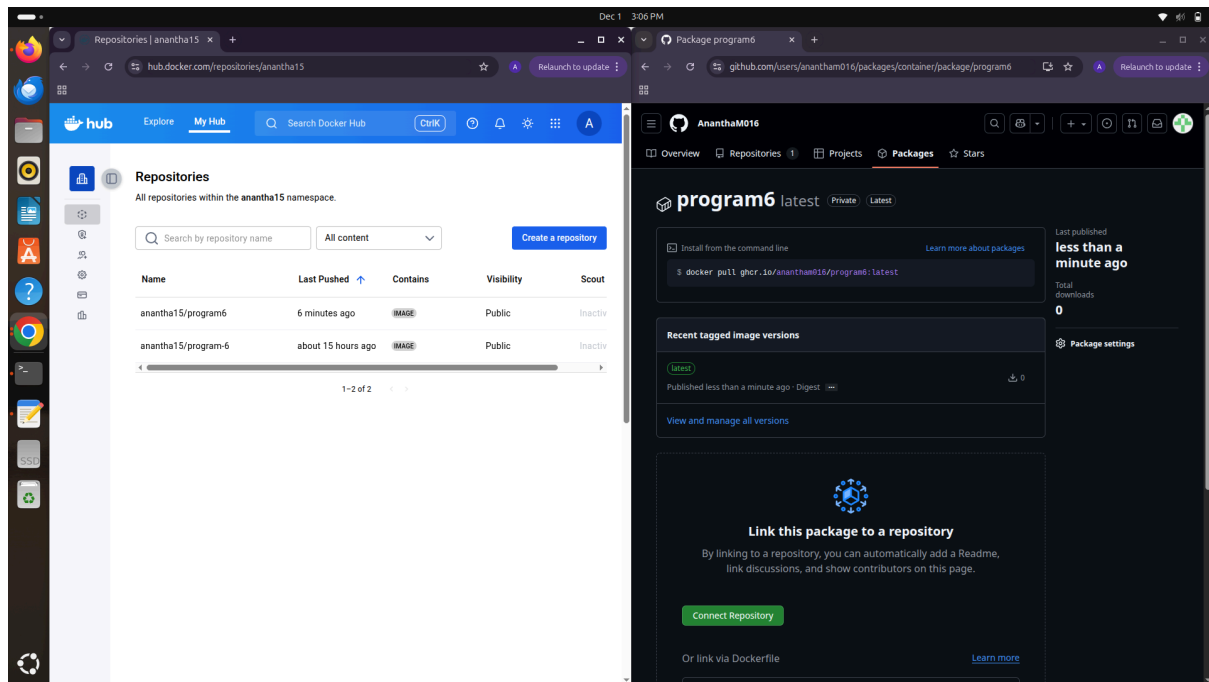
```
#echo <PAT>| docker login ghcr.io -u AnanthaM016 --password-stdin
```

Push the image to ghcr

```
#docker push ghcr.io/anantham016/program6
```

```
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab7
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6$ cd ..
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms$ cd Lab6
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6$ nano Dockerfile
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6$ docker build -t program6 .
[+] Building 3.1s (6/6) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 49B
=> [internal] load metadata for docker.io/library/alpine:latest
=> [auth] library/alpine:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> CACHED [1/1] FROM docker.io/library/alpine:latest@sha256:4b7ce07002c69e0f3d704a9c5d6fd3053be500b7f1c69fc0d08990c2ad0dd412
=> exporting to image
=> => exporting layers
=> => writing image sha256:706db57fb2063f39f69632c5b5c9c439633fda35110e65587c5d8553fd1cc38
=> => naming to docker.io/library/program6
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6$ docker tag program6 anantha15/program6:latest
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6$ docker tag program6 ghcr.io/anantham016/program6:latest
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
anantha15/program6  latest             706db57fb206       7 weeks ago        8.32MB
program6            latest             706db57fb206       7 weeks ago        8.32MB
ghcr.io/anantham016/program6  latest             706db57fb206       7 weeks ago        8.32MB
gcr.io/k8s-minikube/kicbase  v0.0.48            c6b5532e987b       2 months ago       1.31GB
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6$ docker login -u anantha15
Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password:
Login Succeeded
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6$ docker push anantha15/program6
Using default tag: latest
The push refers to repository [docker.io/anantha15/program6]
256f393e029f: Mounted from anantha15/program-6
latest: digest: sha256:9d04ae17046f42ec0cd37d0429ff0edd799d7159242938cc5a964dc38c1b64 size: 527
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6$
```



Create a repository and add file to github

```
#git init
#git add .
#git commit -m "first commit"
#git branch main
#git remote add origin <remote_repository_url>
#git push -u origin Main
```

```
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab6

1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab7 x 1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab6 x v

1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab6$ echo ghp_Tp2rzgha36UketyBbpSrWx1RMgA29L3SM79d | docker login ghcr.io -u AnanthaM016 --password-stdin

WARNING! Your credentials are stored unencrypted in '/home/anantha/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab6$ docker push ghcr.io/anantham016/program6
Using default tag: latest
The push refers to repository [ghcr.io/anantham016/program6]
256f393e829f: Mounted from anantham016/program-6
latest: digest: sha256:9d04ae17046f42ec0cd57d0429fff0edd799d7159242938cc5a964dcd38c1b64 size: 527
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab6$ git init
Initialized empty Git repository in /home/anantha/Desktop/LabPrograms/Lab6/.git/
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab6$ git add .
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab6$ git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/AnanthaM016/program-6.git
git push -u origin main
[main (root-commit) cf43c1a] first commit
1 file changed, 1 insertion(+)
create mode 100644 Dockerfile
Username for 'https://github.com': AnanthaM016
Password for 'https://AnanthaM016@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 229 bytes | 229.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/AnanthaM016/program-6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab6$
```